

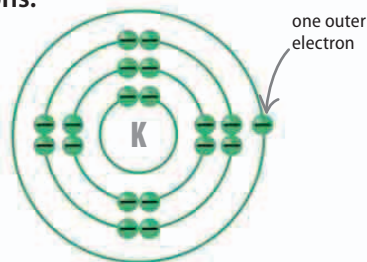
Alkali metals and alkali earth metals

SIX ELEMENTS IN GROUP 1 OF THE PERIODIC TABLE ARE CALLED ALKALI METALS. THE SIX IN GROUP 2 ARE ALKALI EARTH METALS.

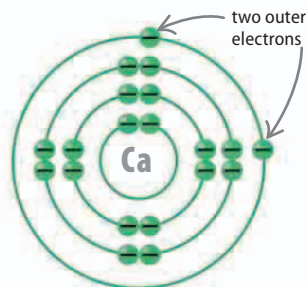
These elements get involved in chemical reactions with other elements easily because they have very few outer electrons.

Reactive metals

Elements in Group 1 have a single outer electron in their atoms, while those of Group 2 have two outer electrons. They form ions easily by losing these electrons, so they readily get involved in reactions, which makes them highly reactive. With just one electron to lose, a member of Group 1, such as potassium (K), will ionize more easily than a member of Group 2, which must lose two electrons.



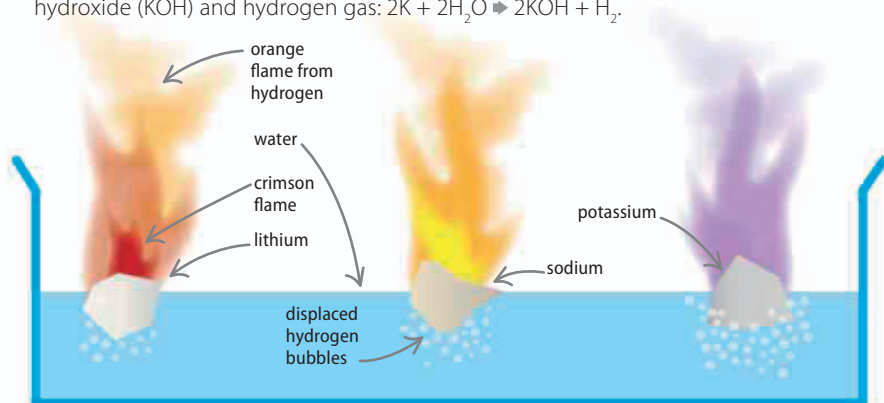
△ **Potassium (Group 1)**
As the third alkali metal, potassium has one electron in its fourth and outermost shell.



△ **Calcium (Group 2)**
Calcium also has four electron shells, but there are two electrons in its outer shell.

Releasing hydrogen

These metals all react strongly with water, producing brightly colored flames. The metal ion swaps places with (displaces) a hydrogen ion in the water, forming a substance called a hydroxide. The displaced hydrogen is released as bubbles of gas. For example, when potassium is added to water, the products are potassium hydroxide (KOH) and hydrogen gas: $2K + 2H_2O \rightarrow 2KOH + H_2$.



△ **Lithium**
When reacting with water, lithium burns with a crimson flame. The hydrogen released turns the flame orange.

△ **Sodium**
This metal produces an orange flame. It is the same color produced by sodium lamps used in street lights.

△ **Potassium**
Potassium burns with a lilac flame. It is more reactive than sodium and lithium and often explodes as it reacts.

REAL WORLD

In bodies

The alkali metals and alkali earth metals are common ingredients in living bodies. Sodium and potassium ions are used to create the electric pulses that fire through muscles and nerves, while calcium compounds are in bones, teeth, and the shells of snails.

